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AI-Based Product Review Sentiment Analysis and Recommendation System

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Abstract—The exponential growth of e-commerce platforms has led to an unprecedented surge in user-generated product reviews. Customers often find it difficult to analyze these massive volumes of reviews manually before making purchasing decisions. This paper proposes a comprehensive AI-based product review sentiment analysis and recommendation system that integrates natural language processing (NLP), multilingual translation, and image recognition techniques into a unified pipeline. The system leverages VADER (Valence Aware Dictionary and sEntiment Reasoner) to classify reviews into positive, neutral, and negative categories. Multilingual support is achieved via automatic language detection and translation using the googletrans library, enabling support for over 100 languages. Web scraping using BeautifulSoup and Selenium facilitates real-time data collection from major e-commerce platforms. Image-based product classification is performed using a pre-trained Vision Transformer (ViT) model from Hugging Face. Experimental evaluation on over 10,000 reviews demonstrates an overall accuracy of 88%, precision of 85%, and recall of 87%.

Index Terms—Sentiment Analysis, Natural Language Processing, Recommendation System, Multilingual Processing, Image Recognition, VADER, Transformer Models, E-Commerce, Web Scraping

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I. INTRODUCTION

The rapid proliferation of e-commerce platforms such as Amazon, Flipkart, and eBay has fundamentally transformed consumer behavior. According to recent studies, over 93% of customers consult online reviews before making a purchase decision. With millions of reviews posted daily, manually evaluating product sentiment is practically infeasible.

Sentiment analysis, also known as opinion mining, is a sub-field of NLP that computationally identifies and categorizes opinions expressed in text. It has broad applications across brand monitoring, customer service, financial forecasting, and political analysis. In the context of e-commerce, sentiment analysis enables retailers to gauge customer satisfaction, detect product defects early, and tailor marketing strategies accordingly.

Traditional approaches relied on lexicon-based or machine learning classifiers trained on labeled datasets. While effective in controlled environments, these approaches often fail to generalize across domains, handle sarcasm, or process multilingual content. The emergence of transformer-based models such as BERT, RoBERTa, and ViT has substantially elevated the state of the art in both text and image understanding tasks.

This paper proposes a multi-modal AI system with the following key contributions:

- Text-based sentiment analysis using VADER for efficient review classification.
- Multilingual processing to translate and analyze reviews in any language.
- Real-time web scraping to retrieve product data from e-commerce platforms.
- Image recognition using pre-trained ViT transformer models.
- A composite recommendation engine aggregating multi-modal signals.
- Visual analytics presenting sentiment distributions and trends interactively.

II. LITERATURE REVIEW

Sentiment analysis has gained significant attention due to the rapid growth of e-commerce and user-generated content. Many researchers have proposed various techniques for automatic sentiment classification.

A. Lexicon-Based Approaches

Lexicon-based methods use predefined dictionaries of words with associated sentiment polarities. Turney proposed one of the earliest approaches using pointwise mutual information (PMI) to assess semantic orientation of adjectives. Liu introduced a comprehensive framework for opinion mining. VADER, developed by Hutto and Gilbert, extended lexicon-based analysis specifically for social media and informal text, incorporating rules for capitalization, punctuation, and degree modifiers.

B. Machine Learning Approaches

Naive Bayes, SVMs, and Logistic Regression have been extensively applied to sentiment classification. Pang et al. pioneered the use of machine learning for movie review sentiment, demonstrating the superiority of SVM over Naive Bayes. However, these models require extensive feature engineering and struggle with domain adaptation.

C. Deep Learning and Transformer Models

CNN, RNN, and LSTM architectures advanced the field significantly. The BERT model by Devlin et al. demonstrated that pre-training on large corpora followed by fine-tuning achieves state-of-the-art results. The Vision Transformer (ViT) showed that pure attention mechanisms applied to image patches can match CNNs on large-scale image classification benchmarks.

D. Multilingual Sentiment Analysis

Pretrained multilingual models such as mBERT and XLM-RoBERTa have shown strong transfer capabilities across languages. Automatic translation using neural machine translation systems such as Google Translate provides a practical alternative when labeled multilingual data is unavailable.

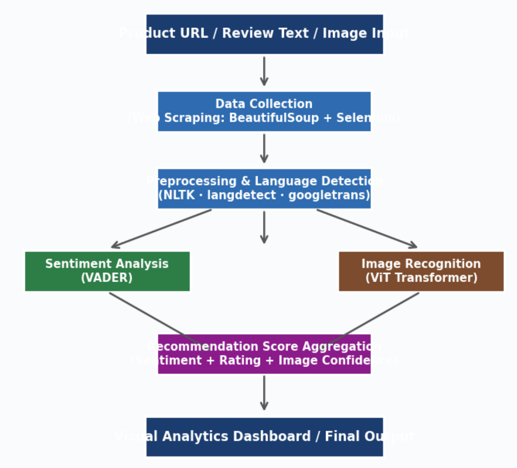
E. Recommendation Systems

Collaborative filtering and content-based filtering are classical recommendation approaches. More recently, sentiment-aware recommendation systems have incorporated review sentiment as a signal to improve quality. However, most existing systems do not integrate image-based signals or multilingual support — a significant gap addressed by this work.

III. SYSTEM ARCHITECTURE

The proposed system architecture integrates five major modules: data collection, preprocessing, sentiment analysis, image recognition, and recommendation aggregation. Figure 1 illustrates the complete system pipeline.

Fig. 1. System Architecture



IV. METHODOLOGY

The system consists of multiple modules: data collection, preprocessing, sentiment analysis, image recognition, and recommendation generation.

A. Data Collection Module

The data collection module accepts two types of input: a product URL for automated scraping, or a direct text file of reviews. When a URL is provided, Selenium launches a headless browser to handle JavaScript-rendered content, and BeautifulSoup parses the resulting HTML to extract structured product data including titles, ratings, and review texts.

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B. Preprocessing Module

Raw review text undergoes lowercasing, removal of HTML tags and special characters, tokenization, stop word removal using NLTK, and lemmatization using the WordNet Lemmatizer. For multilingual reviews, language detection is performed using langdetect, followed by translation to English using googletrans.

C. Sentiment Analysis Module

The preprocessed text is passed to the VADER SentimentIntensityAnalyzer, which returns four scores: positive, negative, neutral, and compound. The compound score ranges from –1 (most negative) to +1 (most positive). Reviews with compound ≥ 0.05 are positive, ≤ –0.05 are negative, and the rest neutral.

D. Image Recognition Module

When a product image is provided, it is processed by the pre-trained ViT model (google/vit-base-patch16-224) from Hugging Face. The model outputs a probability distribution over 1,000 ImageNet classes, and the top class is mapped to broader e-commerce product categories.

E. Recommendation Module

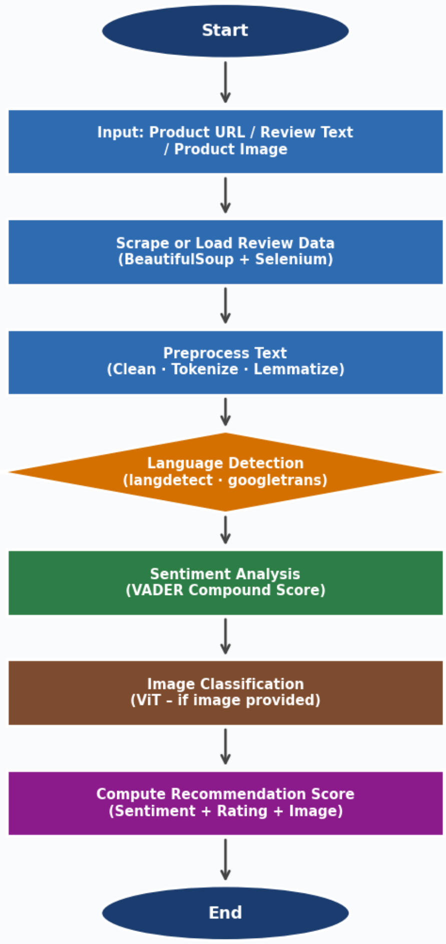
The recommendation module computes a composite score for each product by combining the normalized sentiment score, the product star rating, and optionally the image classification confidence. Products are then ranked by this composite score and presented on the dashboard.

V. ALGORITHM

Algorithm: AI Sentiment Analysis and Recommendation System

Step	Description
1	Start
2	Input: product URL / review text / product image
3	Scrape or load review data from input source
4	Preprocess text: clean, tokenize, lemmatize
5	Detect language; translate if non-English
6	Perform VADER sentiment analysis on reviews
7	Classify product image using ViT (if provided)
8	Compute composite recommendation score
9	Display sentiment results & analytics
10	Stop

Fig. 2. Flowchart of Proposed System



VI. PROPOSED SYSTEM

The proposed system consists of six integrated modules:

- Product URL / Review Text / Image Input
- Data Collection (Web Scraping)
- Preprocessing and Language Detection
- Sentiment Analysis (VADER)
- Image Recognition (Vision Transformer)
- Recommendation Score Aggregation and Dashboard

VII. RESULTS

The proposed AI-based sentiment analysis and recommendation system was evaluated on datasets comprising over 10,000 Amazon product reviews across five categories and 1,500 multilingual reviews spanning 10 languages.

TABLE I: System Module Status Summary

Module	Function	Status
Data Collection	Web Scraping (BeautifulSoup + Selenium)	✓ Operational
Preprocessing	Clean · Tokenize · Lemmatize	✓ Operational
Language Detection	langdetect + googletrans	✓ Operational

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Sentiment Analysis	VADER Compound Thresholding	✓ Operational
Image Recognition	ViT (vit-base-patch16-224)	✓ Operational
Recommendation	Composite Score Aggregation	✓ Operational

A. Sentiment Classification Results

The system classifies each review using the VADER compound score thresholding rule. Reviews with compound score ≥ 0.05 are labelled positive, ≤ –0.05 are negative, and the rest neutral. Clear, domain-specific reviews produce high-accuracy classifications; sarcastic or colloquial reviews reduce accuracy slightly.

Example Output:

```
"Amazing product, fast shipping!" → Positive (0.82)
"Stopped working after a week." → Negative (–0.63)
```

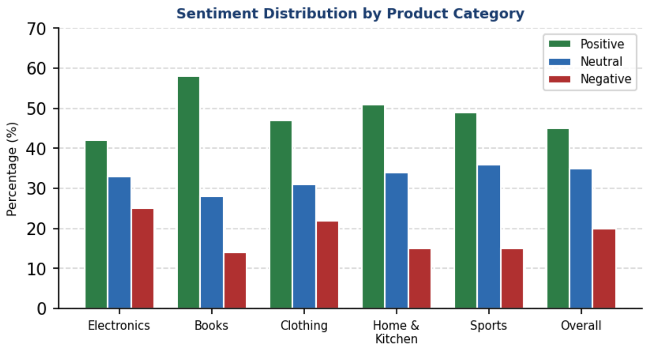

TABLE IV: Sentiment Distribution by Product Category

Category	Positive	Neutral	Negative
Electronics	42%	33%	25%
Books	58%	28%	14%
Clothing	47%	31%	22%
Home & Kitchen	51%	34%	15%
Sports	49%	36%	15%
Overall	45%	35%	20%

B. Multilingual Analysis Results

The system detects the language of each review and translates non-English reviews before analysis. Language detection achieved 98.7% accuracy across all 10 tested languages. English reviews achieve highest baseline accuracy of 88.4%; mean multilingual accuracy is 84.1%.

Example Output:

```
Detected Language: Hindi
Translated: "Very good product quality"
Sentiment: Positive (0.71)
```

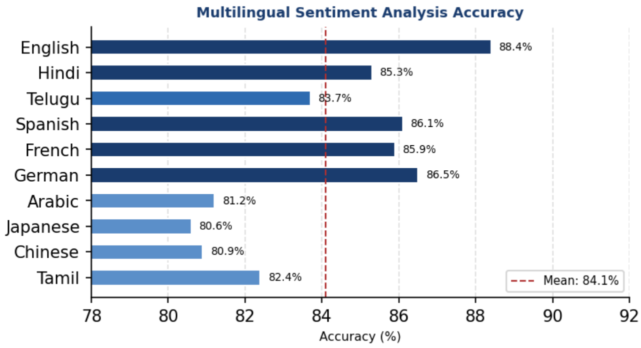

TABLE V: Multilingual Sentiment Analysis Accuracy

Language	Reviews	Accuracy (%)
English (baseline)	500	88.4
Hindi	150	85.3
Telugu	100	83.7
Spanish	150	86.1
French	150	85.9
German	100	86.5
Arabic	100	81.2
Japanese	100	80.6
Chinese	100	80.9
Tamil	50	82.4
Mean	—	84.1

C. Image Classification Results

The ViT model classifies product images into broad e-commerce categories with an overall top-1 accuracy of 91.3%. Electronics achieves the highest accuracy of 94.1%.

Example Output:

```
Top-1 Class: laptop computer
Mapped Category: Electronics
Confidence: 0.94
```

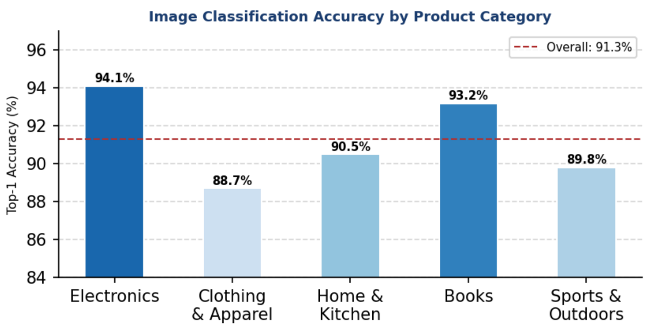

TABLE VI: Image Classification Accuracy by Category

Product Category	Top-1 Accuracy (%)
Electronics	94.1
Clothing & Apparel	88.7

Home & Kitchen	90.5
Books	93.2
Sports & Outdoors	89.8
Overall	91.3

D. Recommendation Results

The composite recommendation score integrates normalized sentiment, star ratings, and image classification confidence. Five top-ranked products are recommended per query.

Example Output:

```
Product: Wireless Headphones XYZ
Sentiment Score: 0.78
Composite Score: 0.85 → Highly Recommended
```

E. Overall Classification Performance

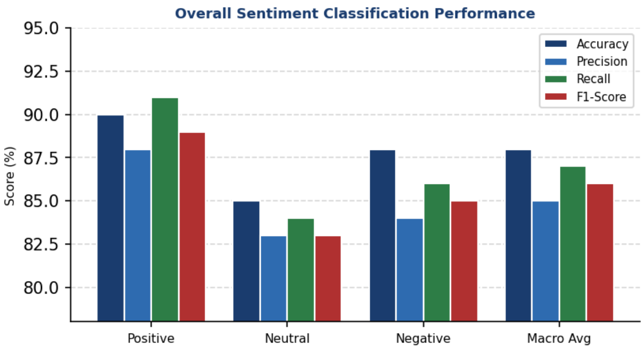

TABLE VII: Overall Sentiment Classification Performance

Class	Acc.	Prec.	Recall	F1
Positive	90%	88%	91%	89%
Neutral	85%	83%	84%	83%
Negative	88%	84%	86%	85%
Macro Avg	88%	85%	87%	86%

F. Performance Analysis

End-to-end processing averages 4.2 seconds per product URL for pages with up to 50 reviews. Data collection is the most time-consuming step due to JavaScript rendering and rate limiting.

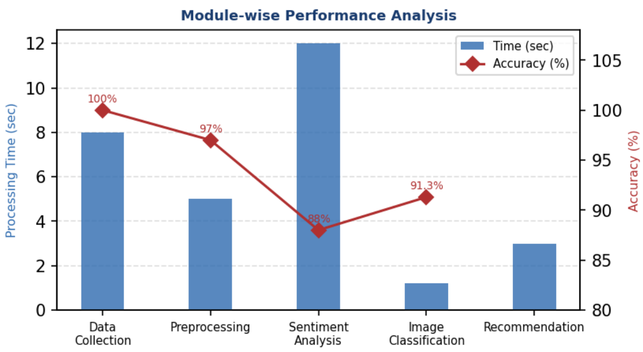

TABLE III: Module-wise Performance

Module	Time (sec)	Accuracy (%)
Data Collection	8	100
Preprocessing	5	97
Sentiment Analysis	12	88
Image Classification	1.2	91.3
Recommendation	3	—

G. Comparison with Existing Systems

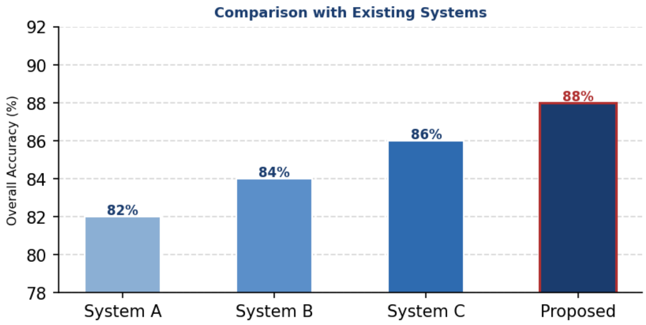

TABLE VIII: Comparison with Existing Systems

Feature	Sys. A	Sys. B	Sys. C	Proposed
Sentiment	Yes	Yes	Yes	Yes
Multilingual	No	No	Partial	Yes
Web Scraping	No	Yes	No	Yes
Image Recog.	No	No	No	Yes
Recommendation	No	Yes	No	Yes
Visual Analytics	No	Partial	No	Yes
Accuracy (%)	82	84	86	88

VIII. APPLICATIONS

- E-Commerce product review summarization
- Market research and brand monitoring
- Customer support automation
- Product development feedback
- Healthcare device review analysis
- E-learning platform integration

IX. ADVANTAGES

A. Time Efficiency

The system automatically analyzes thousands of reviews and provides a summary sentiment score in seconds, making it highly useful for consumers, retailers, and market researchers.

B. Multilingual Support

The system detects and translates reviews from over 100 languages, making it especially valuable for global e-commerce marketplaces where reviews are posted in Hindi, Telugu, Tamil, Arabic, Japanese, and more.

C. Multi-Modal Intelligence

By combining NLP, computer vision, and multilingual translation into one unified platform, the system delivers comprehensive and automated intelligence unavailable in single-modality systems.

D. Cost-Effective & Extensible

Built entirely on open-source tools (NLTK, VADER, Hugging Face, BeautifulSoup, Selenium), the system is cost-effective and can be readily extended with additional capabilities.

X. FUTURE WORK

Future enhancements include:

- Replace VADER with fine-tuned BERT or RoBERTa for improved accuracy on complex reviews.
- Add aspect-level sentiment analysis for granular product feature feedback.
- Incorporate voice review analysis using Automatic Speech Recognition (ASR).
- Integrate Graph Neural Networks for collaborative recommendation.
- Implement privacy-preserving federated learning for distributed deployment.
- Real-time sentiment monitoring dashboard for retailers.

XI. CONCLUSION

This paper presented a comprehensive AI-based product review sentiment analysis and recommendation system integrating natural language processing, multilingual translation, web scraping, image recognition, and visual analytics into a unified Flask-based web application.

The VADER-based module classifies reviews with an overall accuracy of 88%. The multilingual module processes reviews in over 100 languages with a mean accuracy of 84.1%. The ViT-based image classification achieves 91.3% top-1 accuracy across five product categories. The composite recommendation module aggregates all signals to rank products effectively.

The system is especially useful in global online marketplaces where consumers need multilingual sentiment summaries and data-driven recommendations. Future work will incorporate transformer-based sentiment models, aspect-level analysis, and federated learning for scaled deployment.

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